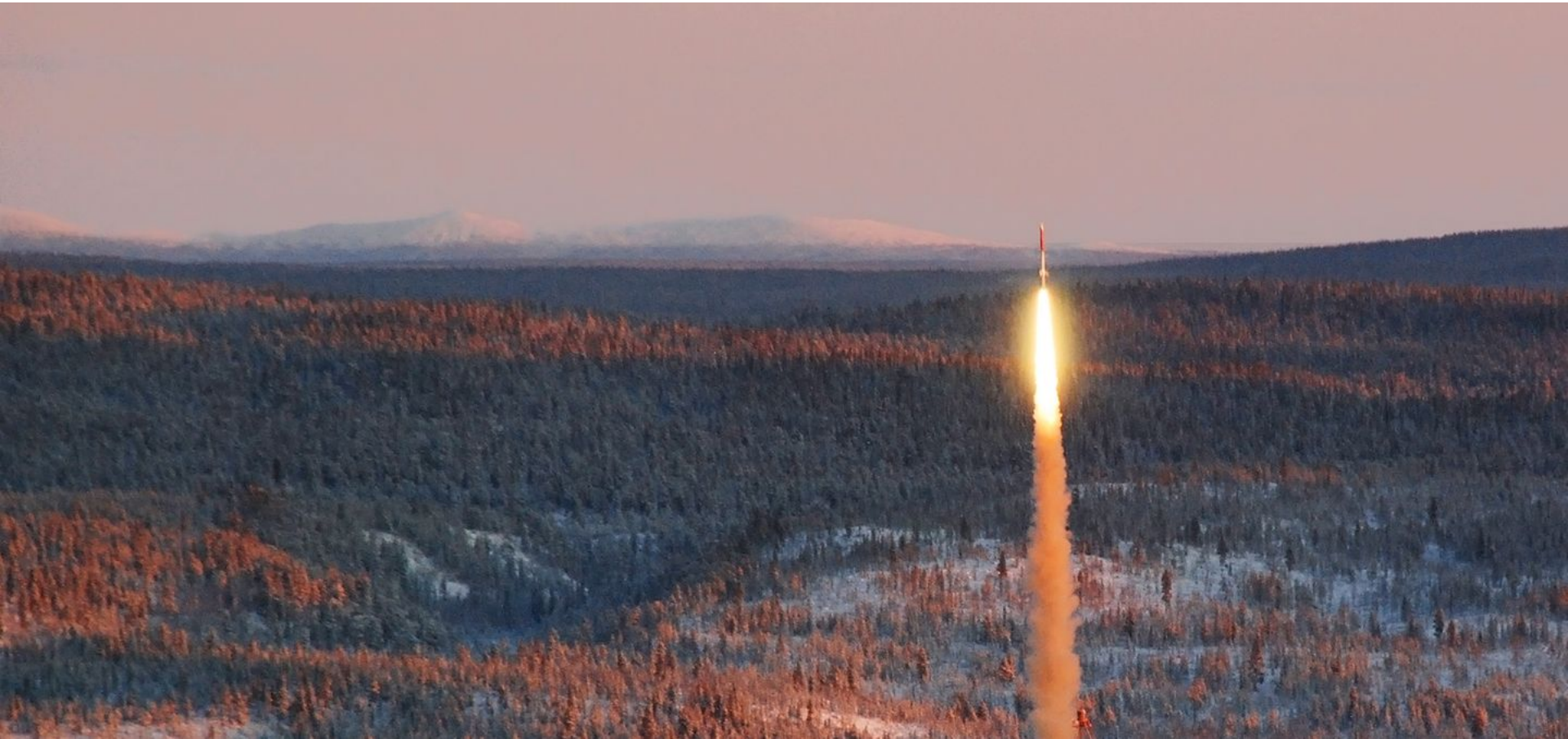


Risk & Uncertainty

42620 Science, Technology & Society

Anders Kristian Munk | Human-Centered Innovation | Department of Technology, Management and Economics





Severity → ↓ Likelihood	Catastrophic	Critical	Marginal	Negligible
Frequent	High	High	Serious	Medium
Probable	High	High	Serious	Medium
Occasional	High	Serious	Serious	Low
Remote	Serious	Medium	Medium	Low
Improbable	Medium	Medium	Medium	Low
Eliminated	Eliminated	Eliminated	Eliminated	Eliminated

RISK

John Adams



◀ **Known unknowns**
Front page of "Risk", 1995

Unknown unknowns ▲
Donald Rumsfeld, 2002

Goal of this lecture: Help you assess if actors actors have different perceptions of risk.

Contribution to your policy brief: Concepts that you can use to analyze actor statements in the energy island controversy and make claims about their underlying risk perceptions and their potential consequences for problem wickedness.

Which of these
two statements
do you agree
with more?

Screens

1 *Hierarchies matter. People have positions in society for a reason: they have different responsibilities and different roles to play. That's important, whether you like it or not.*

2 *In practice, people's formal positions in society are not really important. What matters is their values and their actions, not their jobs, their educations, or their status.*

Foyer

Which of these
two statements
do you agree
with more?

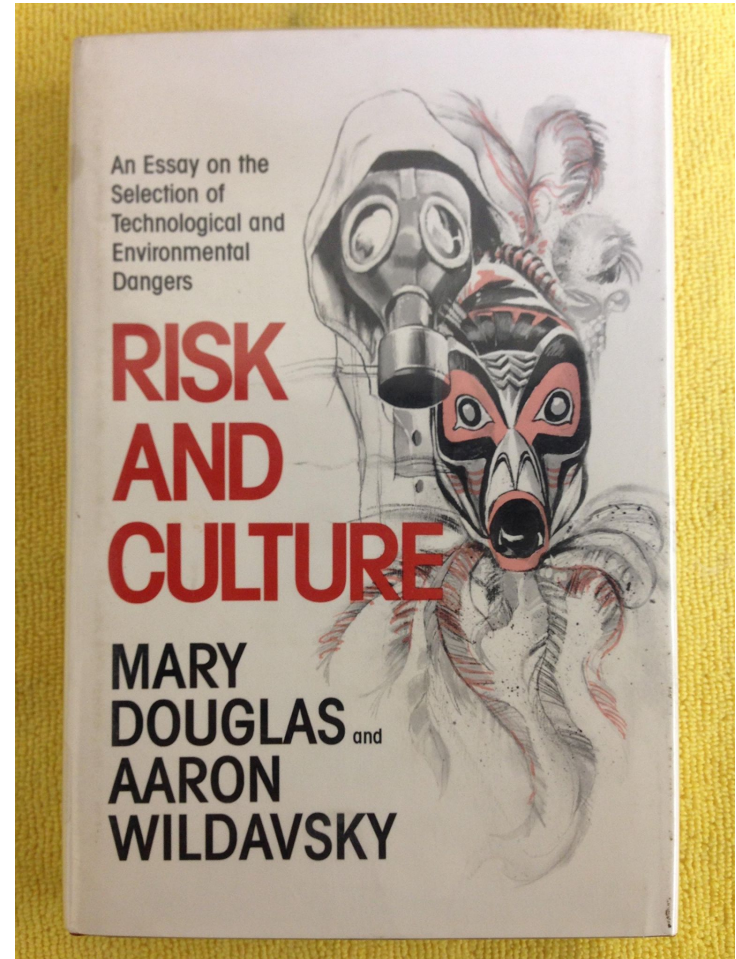
Screens

1 <i>People are individuals. In the end, and whether you like it or not, to be human is to fend for yourself.</i>	2 <i>Communities matter. Being part of a group, whether you like it or not, is one of the most important factors of human life.</i>
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Foyer

Cultural theory of risk

Functionalist anthropology >>> Mary Douglas, 1980s

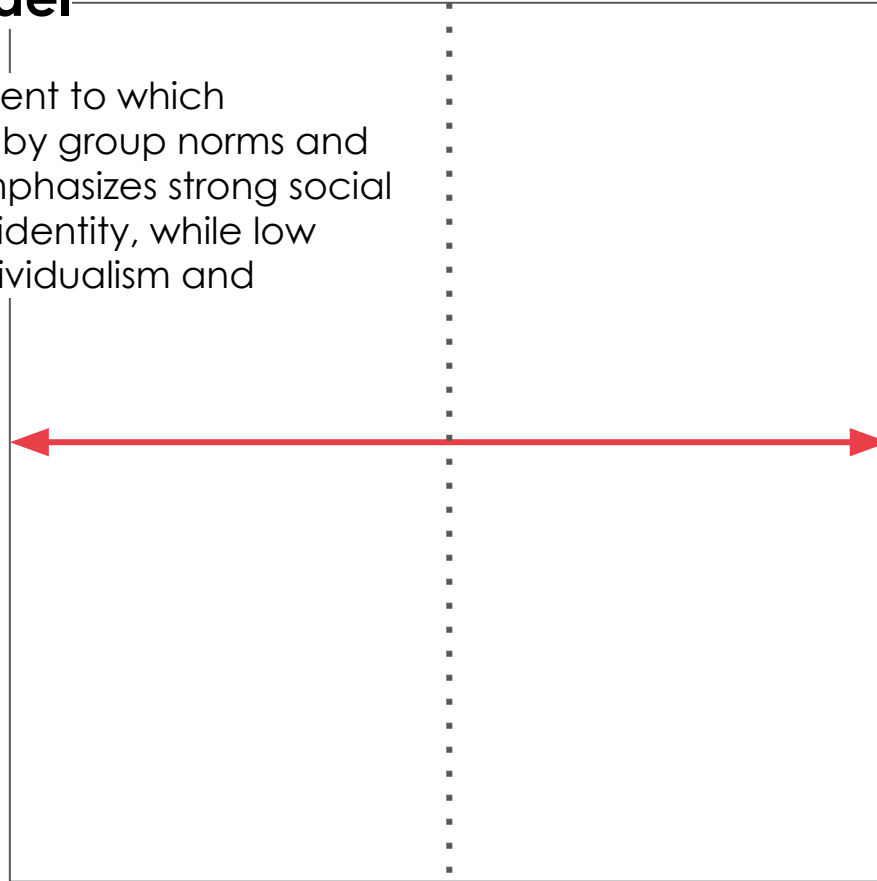


Grid/group model

GROUP dimension: Extent to which individuals are bound by group norms and values. High group emphasizes strong social bonds and collective identity, while low group emphasizes individualism and autonomy.

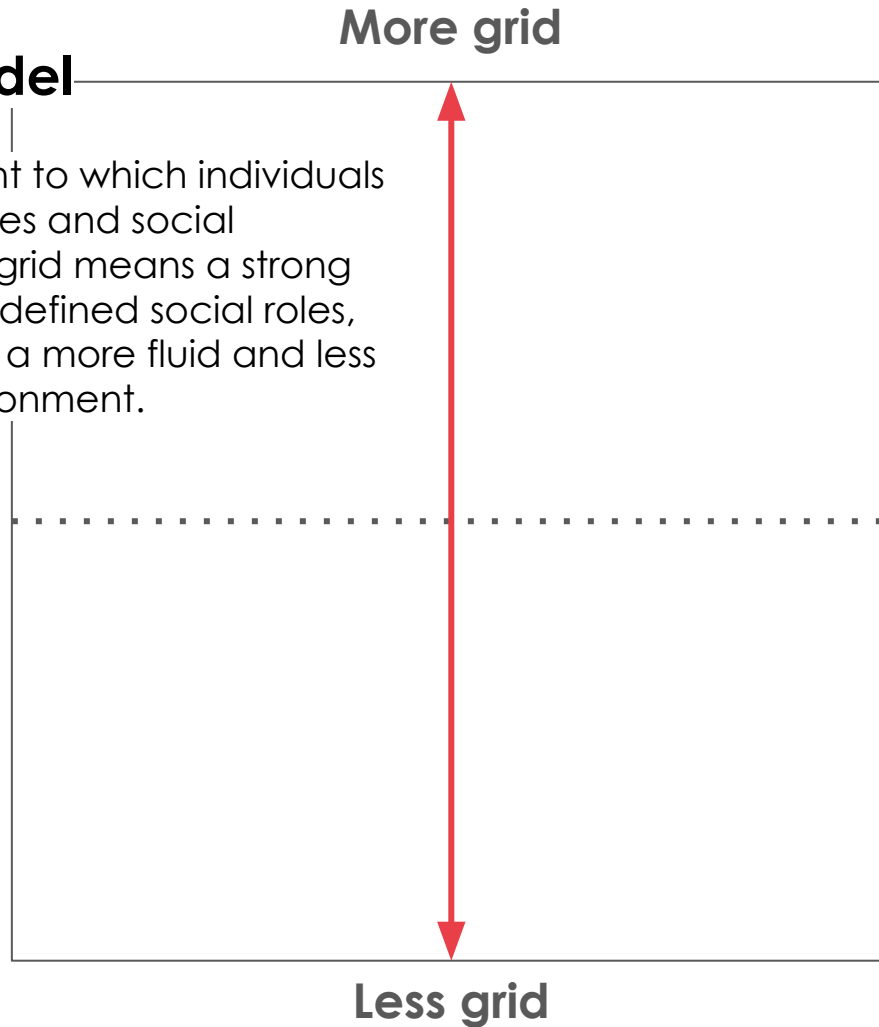
Less group

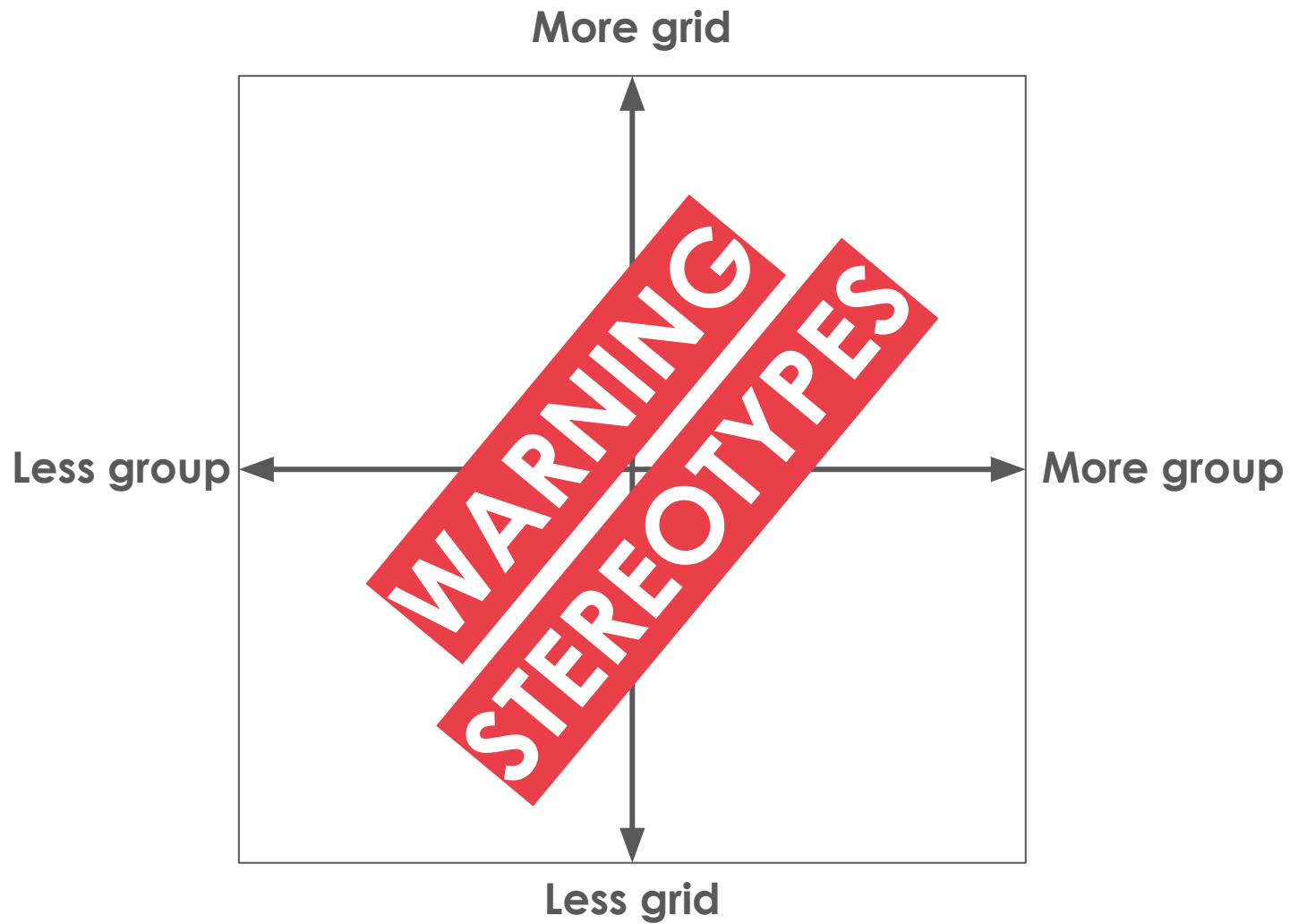
More group



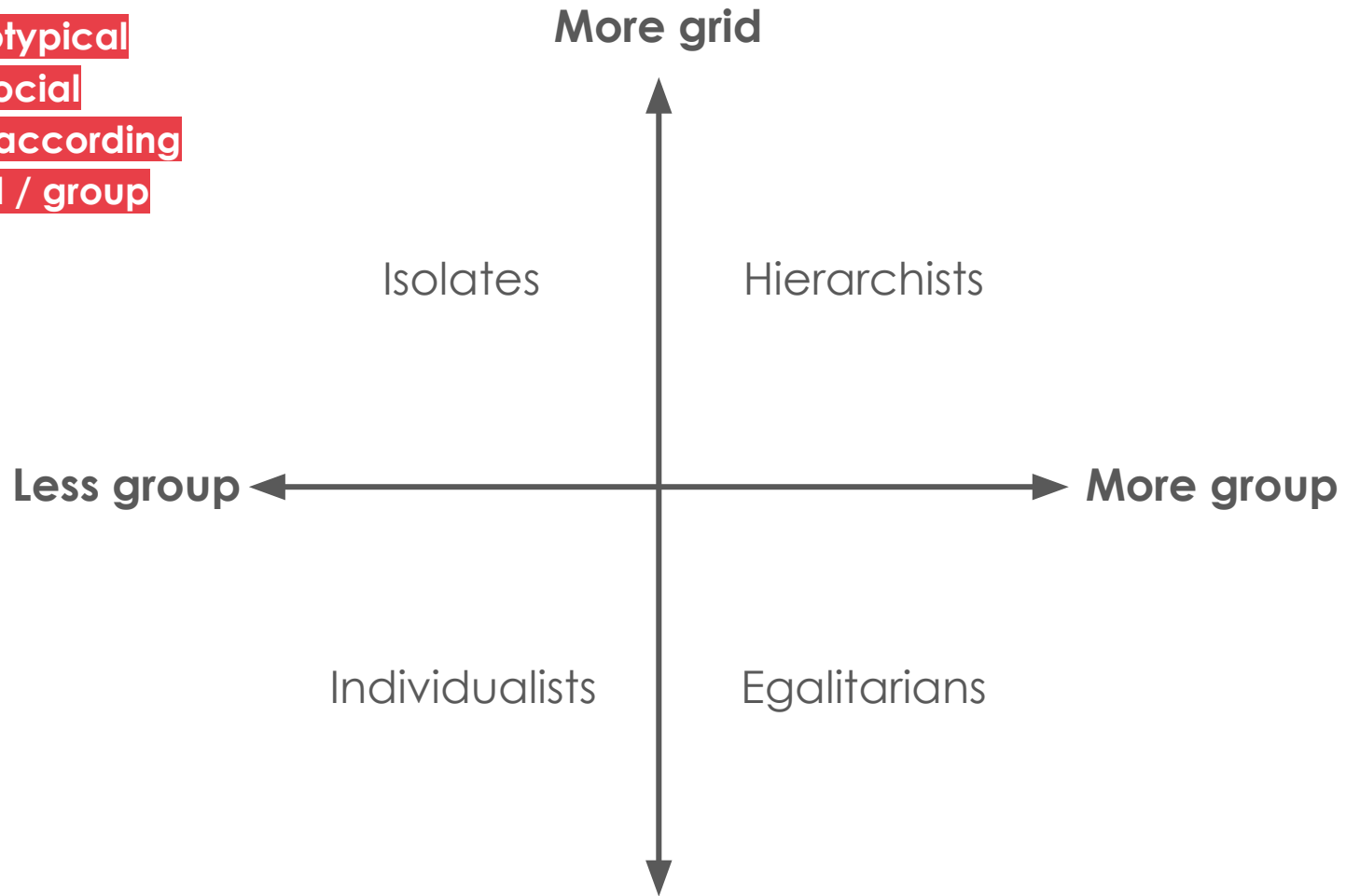
Grid/group model

GRID dimension: Extent to which individuals are constrained by rules and social categorizations. High grid means a strong hierarchy with clearly defined social roles, while low grid signifies a more fluid and less regulated social environment.



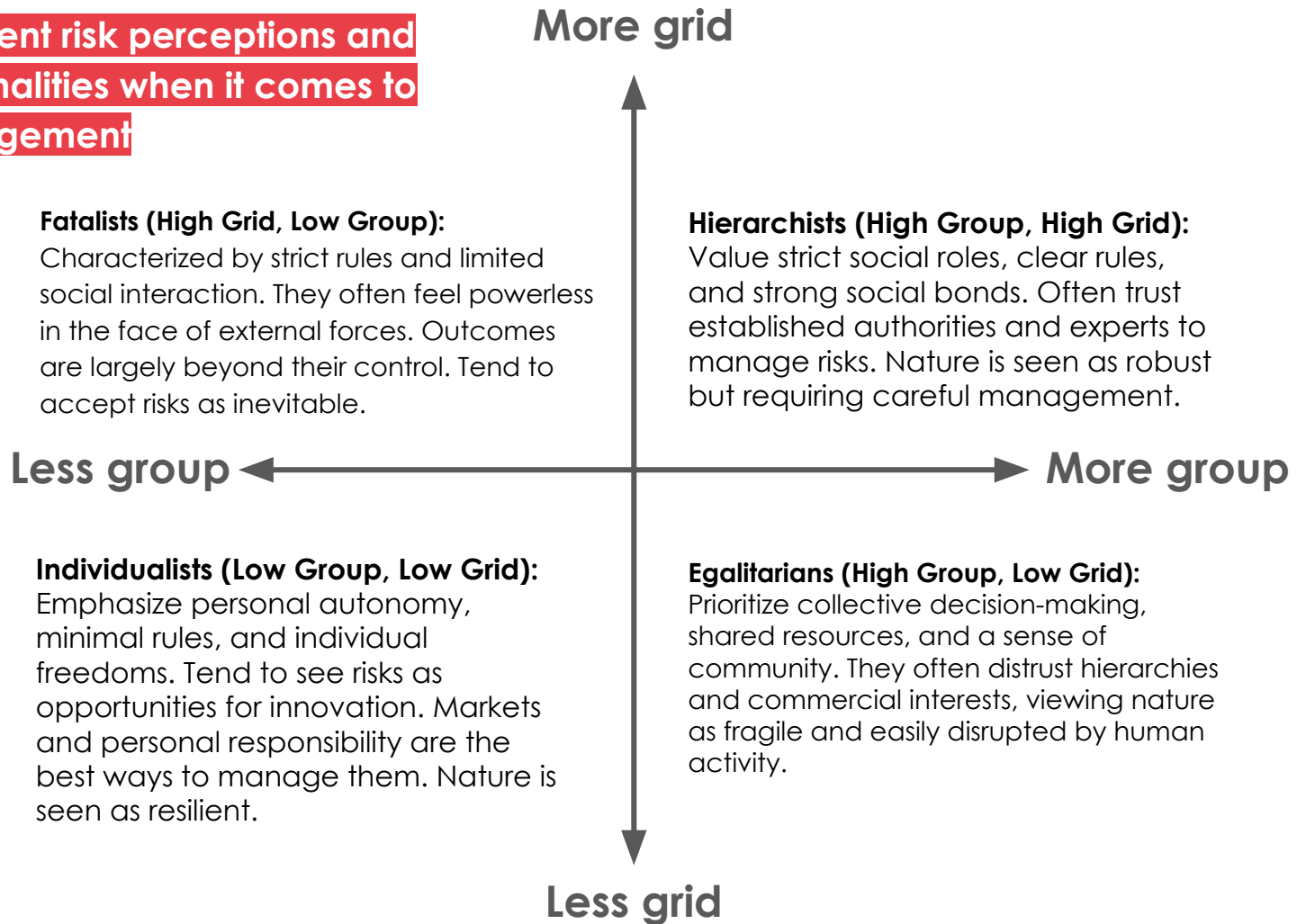


**The stereotypical
forms of social
solidarity according
to the grid / group
model***

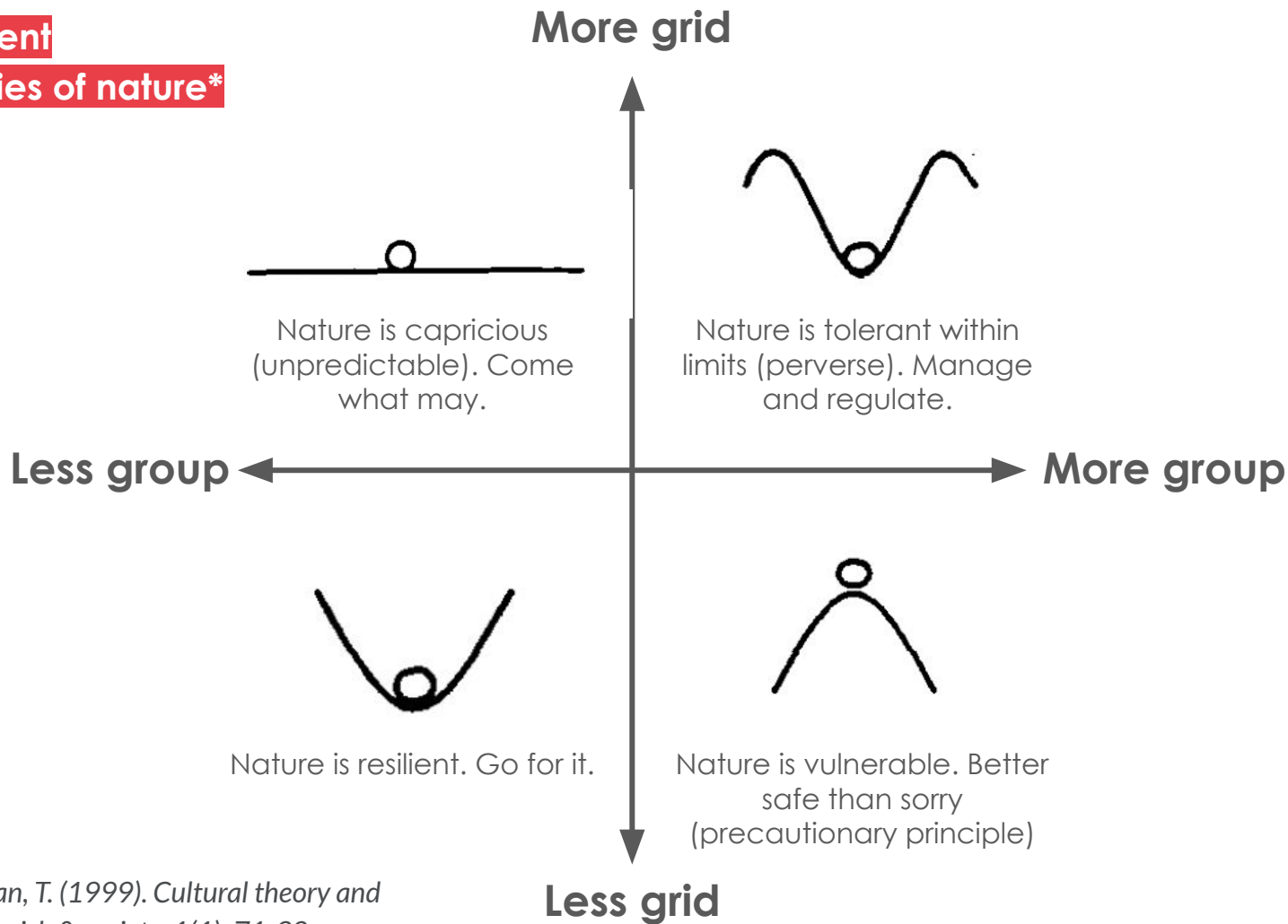


**Tansey, J., & O'riordan, T. (1999). Cultural theory and risk: a review. Health, risk & society, 1(1), 71-90.*

Four different risk perceptions and their rationalities when it comes to risk management

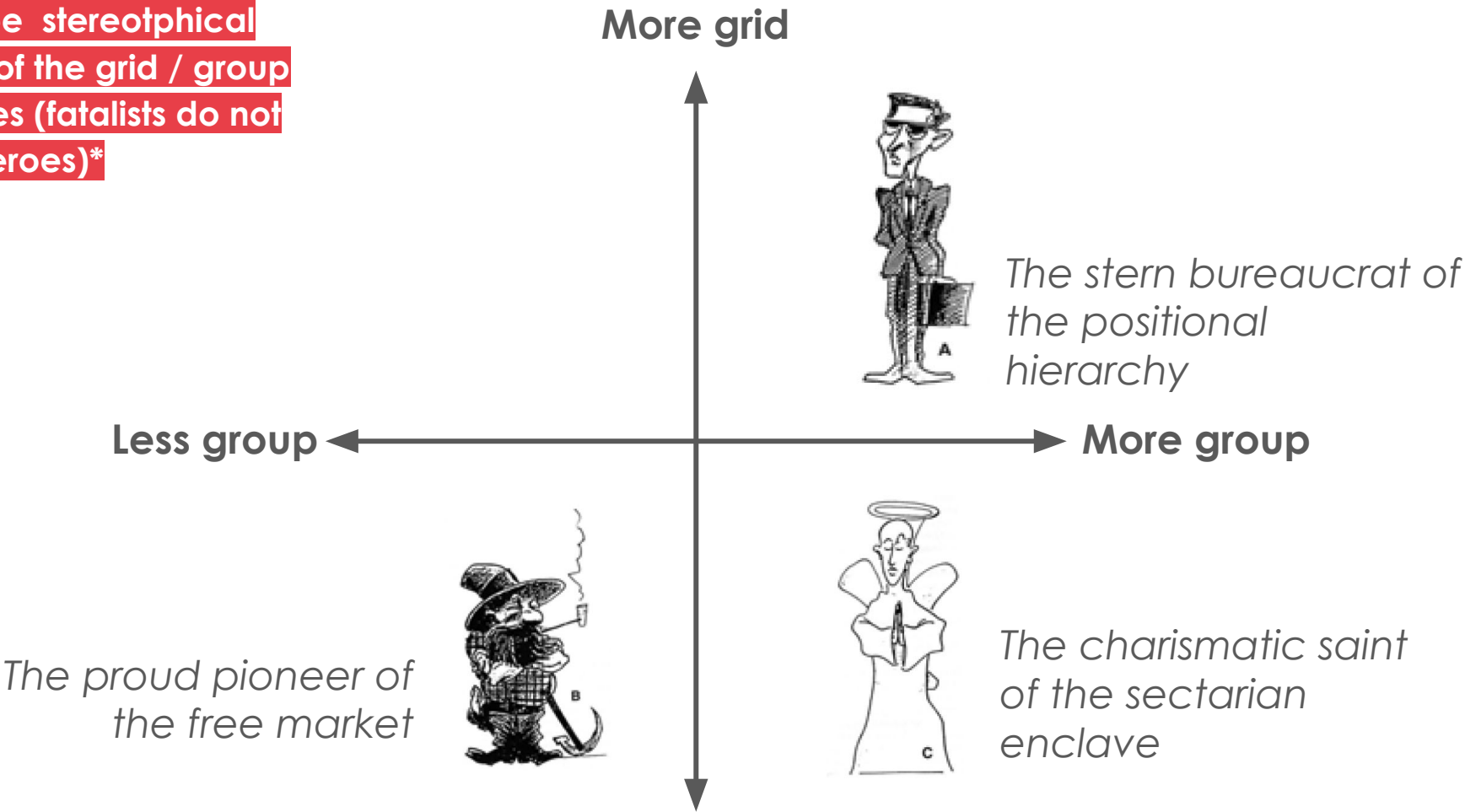


**Four different
philosophies of nature***



*Tansey, J., & O'riordan, T. (1999). Cultural theory and risk: a review. *Health, risk & society*, 1(1), 71-90.

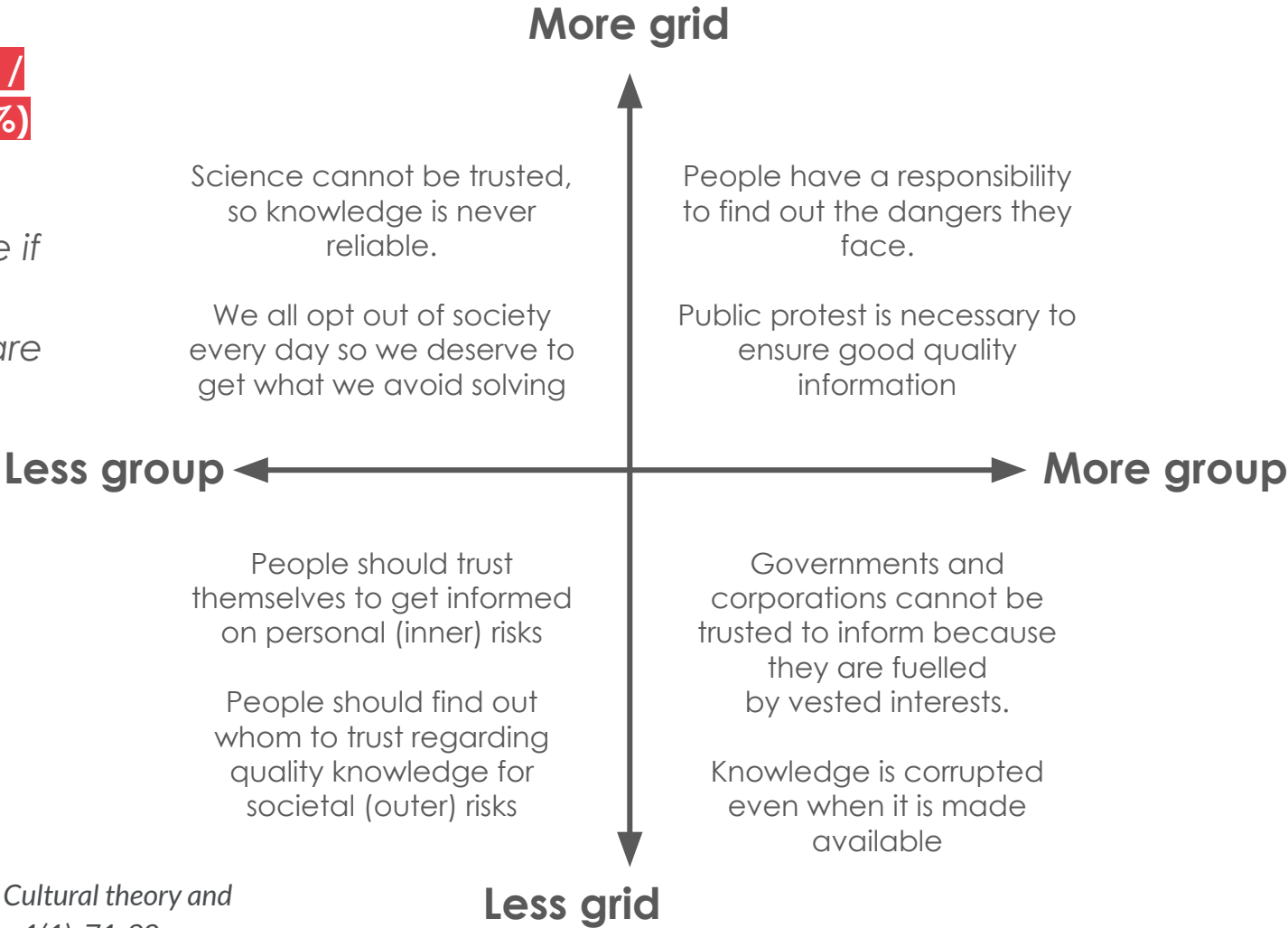
The three stereotypical
heroes of the grid / group
extremes (fatalists do not
have heroes)*



*Douglas, M. (2007). A history of grid and group cultural theory. Toronto, Canada: University of Toronto.

**Survey responses from
people who fit the grid /
group stereotypes (14%)
to the statement:**

*A risk is less acceptable if
knowledge is withheld
from the people who are
exposed to it.*



**Tansey, J., & O'riordan, T. (1999). Cultural theory and risk: a review. Health, risk & society, 1(1), 71-90.*

Label the following position on artificial intelligence according to the cultural theory of risk:



"AI could be a powerful tool for societal efficiency. We should develop it within a framework of ethical guidelines and expert oversight. Important to maintain control and prevent disruption to established systems."



Label the following position on genetic engineering according to the cultural theory of risk:

"GMOs are just another development that we have little control over. We might consume them without much thought. Our fate is largely predetermined regardless of what we eat."





Out of these attitudes to nuclear power risk management, which is the egalitarian (low grid, high group)?

